CODEALPHA INTERNSHIP PROGRAM

Automation System Design

Technical Documentation & Flowchart Implementations

Domain:	Robotics & Automation
Task Reference:	TASK 3 — Flowchart & Circuit Design Specs
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1. Introduction

Modern engineering relies heavily on system automation to maximize efficiency, enhance precision, and minimize human error. This comprehensive technical report fulfills **Task 3: Automation System Design** by presenting formal architectures and control workflows for three essential domains: smart infrastructures, industrial handling, and product packaging.

To ensure production-grade clarity, this document contains descriptive system specifications alongside structured flowcharts for three distinct subtopics:

- **Smart Home Lighting System:** An adaptive control mechanism utilizing human detection and ambient light evaluation to optimize power consumption.
- **Conveyor Belt with Sensors:** An industrial tracking setup designed to isolate defect components while maintaining seamless object handling.
- **Automated Packaging System:** A batch-processing manufacturing routine featuring automated positioning, volumetric container filling, and dynamic box sealing.

2. Subtopic 1: Smart Home Lighting System

2.1 Functional Specification

The Smart Home Lighting System targets energy optimization by decoupling lighting controls from manual operation. The system continually samples data via two main components: a Passive Infrared (PIR) motion sensor to determine room occupancy and a Light Dependent Resistor (LDR) sensor to monitor ambient lux parameters.

The primary controller executes an logical checking process: if human occupancy is recognized, it immediately queries the ambient illumination. If structural illumination drops beneath a threshold (e.g., 300 lux), the central micro-controller acts upon a solid-state relay array to energize the room lighting elements. An inactivity timer prevents flickering; if no movement is recorded for a continuous window of 5 minutes, the lighting matrix turns off safely.

2.2 System Control Logic Flowchart

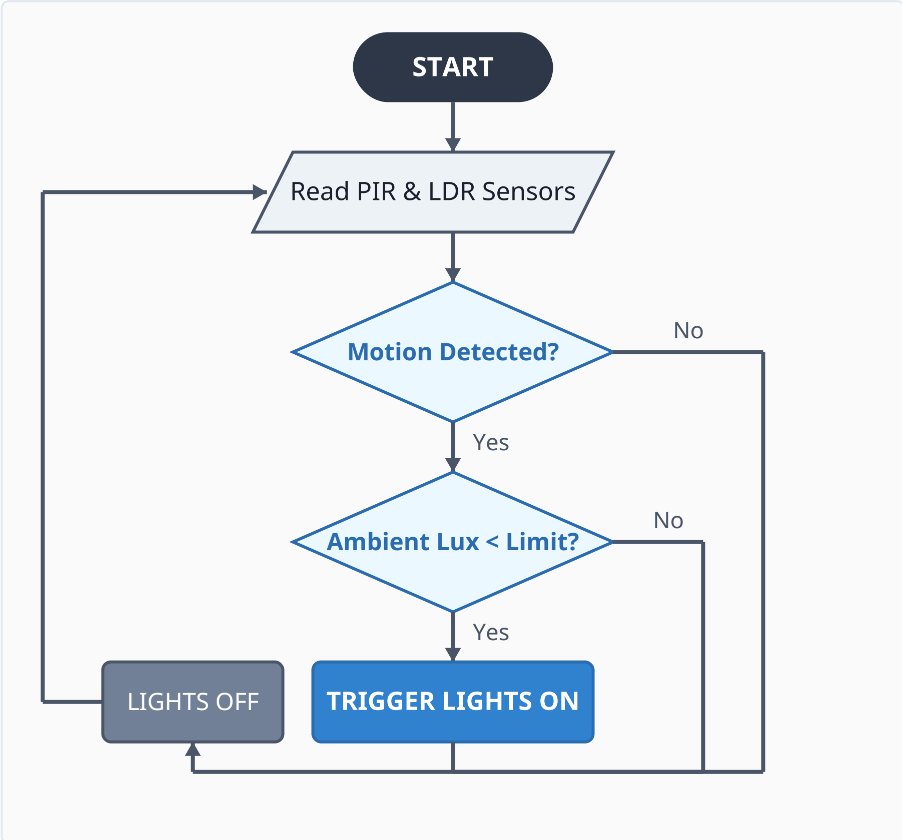

Figure 2.1: Control Logic Loop for the Adaptive Smart Home Lighting Subsystem.

3. Subtopic 2: Conveyor Belt with Sensors (Detailed System Design)

3.1 Functional Specification

The industrial conveyor system automates material handling by performing localized real-time component inspection. Driven by an alternating current (AC) induction motor connected to a variable frequency drive (VFD), the belt moves units sequentially past an inspection terminal. This terminal integrates two active data collectors: a photoelectric proximity sensor that tallies the product influx and an optical color/profile sensor that scans for defect indications.

When a structural defect is isolated, the internal program correlates its structural coordinates with encoder counts, allowing the product to progress up to a downstream rejection station. Once the item is aligned with the sorting mechanism, an ultra-fast pneumatic actuator valve triggers, generating a localized burst of compressed air to push the out-of-spec unit into a secondary containment bin without disturbing downstream product flow.

3.2 System Hardware Architecture

Component	Functional Specification	Operational Duty Description
Programmable Logic Controller (PLC)	Siemens S7-1200 CPU	Central computing node evaluating high-speed interrupts from proximity sensors and driving output coils.
Proximity Sensor	Infrared Photoelectric	Emits focused light beams to verify presence and increment overall production throughput counters.
Defect Sensor	Vision / Color Sensor	Performs pixel-level verification against a set structural model to detect surface cracks or wrong dimensions.
Pneumatic Actuator	Double-Acting Cylinder	Deploys a high-velocity mechanical stroke to redirect out-of-spec parts upon receiving a 24V signal.

3.3 Conveyor System Control Circuit Diagram

The following electrical diagram demonstrates the control interface layout. It maps the integration of sensors and high-voltage power components with the central processing unit:

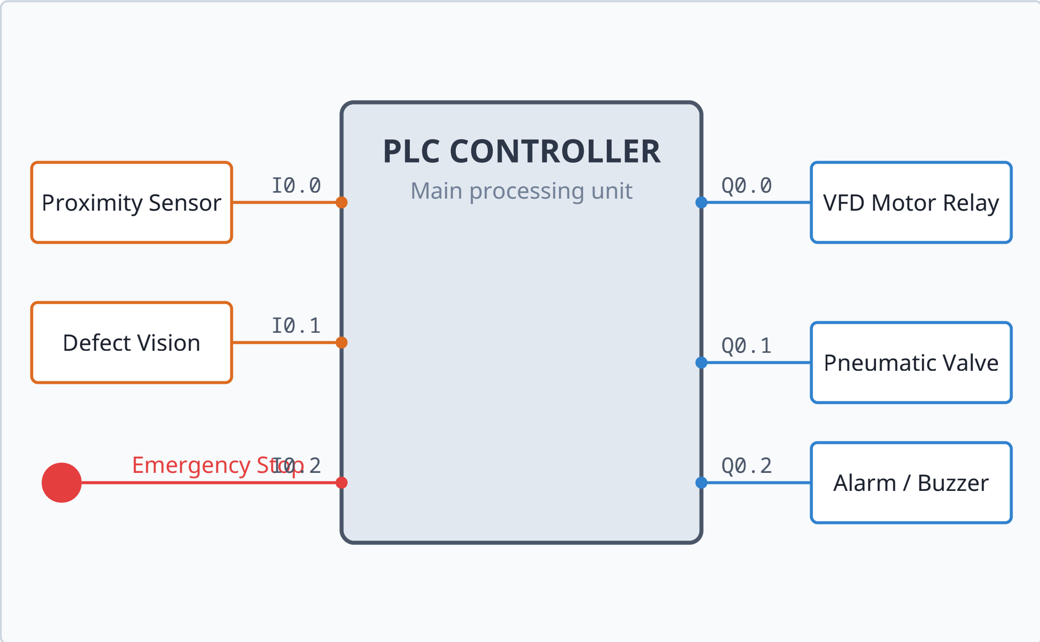

Figure 3.1: Hardware Interface I/O Schematics for the Conveyor Sorting Controller.

3.4 Conveyor Sorting Process Flowchart

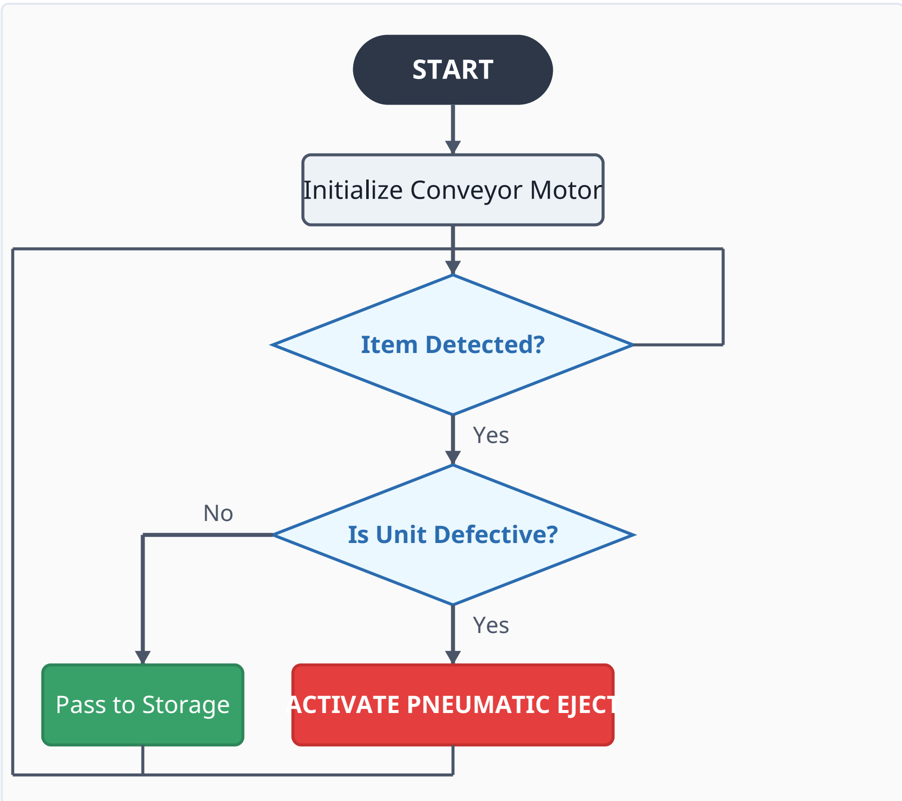

Figure 3.2: Algorithmic Logic Flow Chart for Automated In-line Rejection.

4. Subtopic 3: Automated Packaging System

4.1 Functional Specification

The Automated Packaging System brings together discrete manufacturing outputs into shipping-ready containers. The packaging procedure runs over three mechanical zones:

1. **Positioning Stage:** A proximity indexing sensor detects an empty packaging box underneath the primary volumetric discharge funnel and brings the primary conveyor belt to a temporary halt.
2. **Filling Stage:** The PLC switches on an electromechanical filling valve. A load-cell transducer continuously monitors weight variations beneath the box. Once the product payload matches the structural parameters, the filling valve shuts down.
3. **Sealing Stage:** The conveyor moves the filled package to a closing thermal press station. Upper pneumatic clamps apply specific pressure and heat to seal the box protective flaps, completing the packaging run.

> **Safety Loop Watchdog:** A continuous safety loop runs concurrently with the logic flow. If structural errors occur—such as a container jam or material shortage—the controller halts processing and turns on an audible alarm tower.

4.2 Packaging System Sequence Flowchart

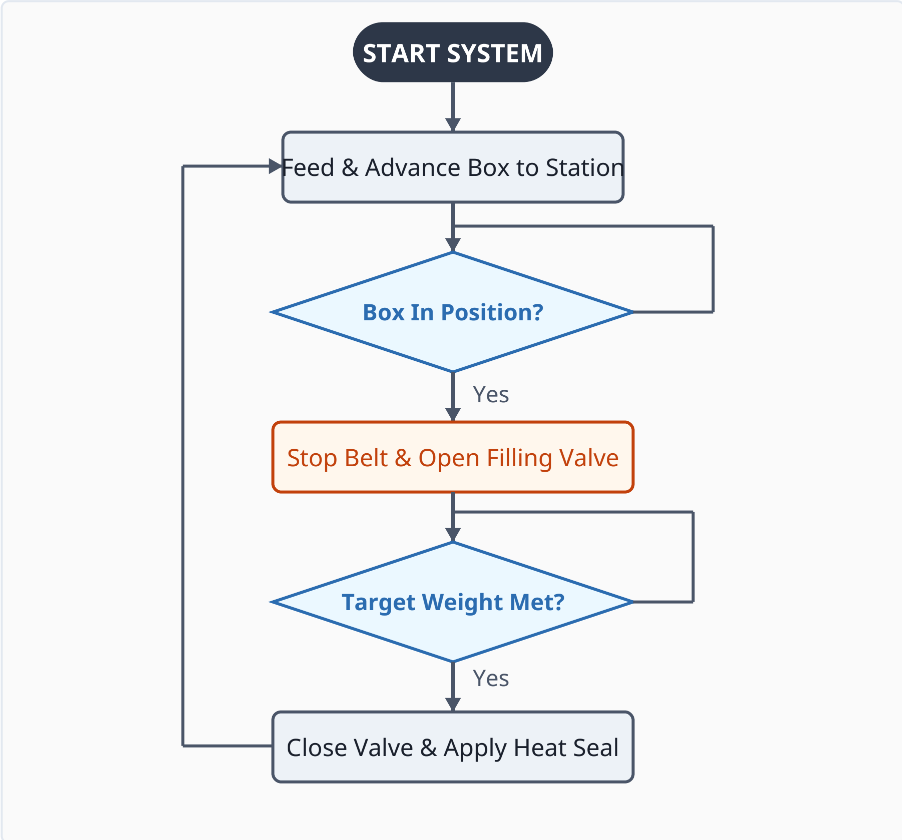

Figure 4.1: Operational Sequence for Box Positioning, Volumetric Filling, and Thermal Sealing.

5. Summary of System Integration

The three industrial setups detailed in this report represent the core building blocks of modern automated factories. By migrating from isolated, hard-wired configurations to flexible, software-driven industrial architectures (such as the Siemens PLC ecosystem), factories can easily adapt their production outputs to changing operational requirements without expensive hardware overalls.

Using continuous sensor feedback loops guarantees strict quality controls across every stage of production. This drastically drops material waste, prevents downstream sorting issues, and helps production environments maintain high throughput and safety standards.